Cambridge O Level Computer Science

Chapter 10: Boolean logic

Lecture notes with Check-Yourself questions and a full answer key.

In this chapter you will learn about:

- The identification, definition, symbols and functions of the standard logic gates: NOT, AND, OR, NAND, NOR and XOR.
- How to use logic gates to create logic circuits from a given problem, a logic expression or a truth table.
- How to complete truth tables from a given problem, a logic expression or a logic circuit.
- How to write a logic expression from a given problem, a logic circuit or a truth table.

Electronic circuits in computers, solid state drives and controlling devices are made of thousands of logic gates. Logic gates take binary inputs and produce a binary output; several gates combined form a logic circuit designed for a specific function, such as controlling a robot. Truth tables check the output of a gate or circuit. Boolean algebra notation is included for completeness but is optional at IGCSE.

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10.1 Standard logic gate symbols

Six logic gates are used in this chapter: NOT, AND, OR, NAND, NOR and XOR. The NOT gate is the only one with a single input; the other five have two inputs. A logic circuit can have more than two inputs overall.

A truth table lists every possible binary input combination and the output for each. The number of combinations is 2 raised to the power of the number of inputs: 2 inputs give 4 combinations, 3 inputs give 8, and 4 inputs give 16.

Standard input orderings (memorise these patterns):

Two inputs (A, B)	**Three inputs (A, B, C)**
0 0	0 0 0
0 1	0 0 1
1 0	0 1 0
1 1	0 1 1
	1 0 0

	1 0 1
	1 1 0
	1 1 1

Check yourself

1. 1. Name the six standard logic gates.
2. 2. Which logic gate has only one input?
3. 3. How many binary combinations are there for a truth table with five inputs? Show your working.
4. 4. Write out all eight input combinations for a three-input truth table.

10.2 The function of the six logic gates

Each gate's function, truth table and notation (logic notation plus optional Boolean algebra, where . means AND, + means OR and a bar over a letter means NOT):

Gate	Output X is 1 if	Truth table (A B -> X)	How to write this
NOT	The input A is 0	0 -> 1; 1 -> 0	X = NOT A
AND	Both inputs A and B are 1	00->0, 01->0, 10->0, 11->1	X = A AND B (Boolean: X = A.B)
OR	Either input A or B, or both, is 1	00->0, 01->1, 10->1, 11->1	X = A OR B (Boolean: X = A + B)
NAND (NOT AND)	Inputs A and B are NOT both 1	00->1, 01->1, 10->1, 11->0	X = A NAND B (Boolean: X = NOT(A.B))
NOR (NOT OR)	Neither input A nor input B is 1	00->1, 01->0, 10->0, 11->0	X = A NOR B (Boolean: X = NOT(A + B))
XOR	Exactly one input is 1: (A is 1 AND B is 0) OR (A is 0 AND B is 1)	00->0, 01->1, 10->1, 11->0	X = A XOR B

The XOR gate in Boolean algebra: X = (A AND NOT B) OR (NOT A AND B), sometimes written X = (A + B) AND NOT(A AND B).

Worked example: Show that X = (A AND NOT B) OR (NOT A AND B) and Y = (A OR B) AND (NOT (A AND B)) both describe the XOR gate.

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Evaluate both for all four input pairs.

A=0, B=0: X = (0 AND 1) OR (1 AND 0) = 0 OR 0 = 0. Y = (0 OR 0) AND NOT(0 AND 0) = 0 AND 1 = 0.

A=0, B=1: X = (0 AND 0) OR (1 AND 1) = 0 OR 1 = 1. Y = (0 OR 1) AND NOT(0 AND 1) = 1 AND 1 = 1.

A=1, B=0: X = (1 AND 1) OR (0 AND 0) = 1 OR 0 = 1. Y = (1 OR 0) AND NOT(1 AND 0) = 1 AND 1 = 1.

A=1, B=1: X = (1 AND 0) OR (0 AND 1) = 0 OR 0 = 0. Y = (1 OR 1) AND NOT(1 AND 1) = 1 AND 0 = 0.

Both give 0, 1, 1, 0: exactly the XOR truth table.

Check yourself

5. 1. State in words when the output of each of the six gates is 1.
6. 2. Write the truth tables for the NAND, NOR and XOR gates.
7. 3. What do the Boolean symbols . , + and a bar over a letter mean?
8. 4. How is NAND related to AND? How is NOR related to OR?

10.3 Logic circuits, logic expressions, truth tables and problem statements

Gates combined to carry out a function form a logic circuit. You must be able to move between four representations: problem statement, logic circuit, logic expression and truth table, in every direction. The method: split a circuit into stages, work out each stage's intermediate values, then combine them.

Example 1: truth table from a logic circuit

Circuit with inputs A, B, C. Stage 1: P = A AND B; Q = B NOR C. Stage 2: R = P XOR Q. Stage 3: X = R XOR C. Work through all eight input combinations, keeping the intermediate columns P, Q and R (leave them in until you are confident):

A	B	C	P (A AND B)	Q (B NOR C)	R (P XOR Q)	X (R XOR C)

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0	0	0	0	1	1	1
0	0	1	0	0	0	1
0	1	0	0	0	0	0
0	1	1	0	0	0	1
1	0	0	0	1	1	1
1	0	1	0	0	0	1
1	1	0	1	0	1	1
1	1	1	1	0	1	0

Final truth table (intermediates removed): X = 1, 1, 0, 1, 1, 1, 1, 0 for the eight input rows in standard order. Check one row by hand: A=1, B=1, C=1 gives P=1, Q=0 (1 NOR 1 = 0), R=1 (1 XOR 0), X=0 (1 XOR 1). Correct.

Example 2: logic expressions from logic circuits

Work from the gates nearest the inputs towards the output, joining each stage's expression with the next gate:

- (a) Gate 1: (A AND B); gate 2: (B OR C); gate 3 joins them with XOR: X = (A AND B) XOR (B OR C).
- (b) Gate 1: (A NAND C); gate 2: (B AND C); gate 3: (gate 1) NOR A, giving ((A NAND C) NOR A); gate 4 joins with OR: X = ((A NAND C) NOR A) OR (B AND C).

Square brackets may be added for clarity but are not required.

Example 3: logic circuit and truth table from a logic expression

Expression: (A XOR C) OR (NOT C NAND B). The connecting gate is OR. Build the left half from (A XOR C) and the right half from (NOT C NAND B), then join with OR. Stage values: P = A XOR C; Q = NOT C; R = Q NAND B; X = P OR R:

A	**B**	**C**	**P (A XOR C)**	**Q (NOT C)**	**R (Q NAND B)**	**X (P OR R)**
0	0	0	0	1	1	1
0	0	1	1	0	1	1
0	1	0	0	1	0	0
0	1	1	1	0	1	1
1	0	0	1	1	1	1
1	0	1	0	0	1	1
1	1	0	1	1	0	1
1	1	1	0	0	1	1

Check yourself

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